INDEX

INSTITUTE

VISION MISSION

INSTITUTE
VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning..

DEPARTMENT

VISION MISSION

DEPARTMENT VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

PROGRAM EDUCATIONAL OBJECTIVES, PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations**.** **[K3]**
6. **The Engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication**: Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance**: Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

Grams: "TECHNOLOGY"
Email: dapjntuk@gmail.com

Phone: 0884-2300991
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Directorate of Academic & Planning
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA
KAKINADA-533003, Andhra Pradesh, INDIA
(Established by AP Government Act No. 30 of 2008)

Lr. No. JNTUK/DAP/AC/B. Tech/IV Year/2019-20 *Date: 30-05-2019*

Dr. A. Mallikarjuna Prasad
M.E, Ph.D.,
Director, Academic Planning

To
All the Principals of Affiliated Colleges,
JNTUK, Kakinada

ACADEMIC CALENDAR FOR B.TECH IV YEAR (2016 BATCH)

I SEMESTER			
Description	**From**	**To**	**Weeks**
Commencement of Class Work	**10.06.2019**		
I Unit of Instructions	10.06.2019	03.08.2019	8W
I Mid Examinations	05.08.2019	10.08.2019	1W
II Unit of Instructions	12.08.2019	05.10.2019	8W
II Mid Examinations	07.10.2019	12.10.2019	1W
Preparation & Practicals	14.10.2019	19.10.2019	1W
End Examinations	21.10.2019	02.11.2019	2W
Commencement of II Semester Class Work	**18.11.2019**		
II SEMESTER			
I Unit of Instructions	**18.11.2019**	11.01.2020	8W
I Mid Examinations	13.01.2020	23.01.2020	1W
II Unit of Instructions	24.01.2020	21.03.2020	8W
II Mid Examinations	23.03.2020	28-03-2020	1W
Preparation	30.03.2020	04.04.2020	1W
End Examinations	06.04.2020	18.04.2020	2W

A.m.prasad
Director Academic Planning

Copy to the Secretary to the Hon'ble Vice Chancellor, JNTUK.
Copy to PA to the Rector, JNTUK.
Copy to PA to the Registrar, JNTUK.
Copy to PA to the Director of Evaluation, JNTUK.

CO-CURRICULAR AND EXTRA CURRICULAR ACTIVITIES

05/06/2019 – World Environment Day	**15/09/2019 – Engineers Day**		**21/03/2019 – International Forest Day**
21/08/2019 to 24/08/2019– 1st year B.Tech. Introduction program	**16/09/2019 – World Ozone Day**	**22/12/2019 – National Mathematics Day**	**22/03/2019 –World Water Day**
15/08/2018 – Independence Day	**September – Intramurals**	**26/01/2019 – Republic Day**	**23/03/2019 – World Meteorological Day**
05/09/2019 – Teachers Day	**11/11/2019 – National Education Day**	**08/03/2019 – International Women's Day**	**24/03/2019 – Earth Hour**
	03/12/2019 – Antipollution Day		**In the month of March – Association Days**

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science &Engineering (Accredited by NBA)

CLASS CONSOLIDATED TIME TABLE

Class: IV B.Tech II Semester — w.e.f. 18-11-2019

Sec : A — Class Coordinator : **Mrs. D. SasiRekha** — Room No : PG-202

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	OT	MS	ML	DS	LUNCH BREAK	CPP	DS	**LIB/SPROTS**
Tue	Seminar	**Project Lab**				ML	DS	MS
Wed	MS	DS	CPP	ML		**Project Lab**		
Thu	OT	MS	CPP	ML		CPP	**Seminar**	
Fri	CPP	**Project Lab**				MS	DS	ML

Class: IV B.Tech II Semester — w.e.f. 18-11-2019

Sec : B — Class Coordinator : **Mr. B Krishna Prasad** — Room No : PG-203

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	DS	**Project Lab**			LUNCH BREAK	CPP	ML	MS
Tue	OT	CPP	MS	DS		**Project Lab**		
Wed	ML	**Seminar**	CPP	ML		MS	DS	**LIB/SPROTS**
Thu	CPP	DS	ML	MS		**Project Lab**		
Fri	OT	MS	DS	CPP		ML	**Seminar**	

Class: IV B.Tech II Semester — w.e.f. 18-11-2019

Sec : C — Class Coordinator : **Mrs. D S L Manikanteswari** — Room No : PG-204

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	CPP	**Project Lab @EF CODD**			LUNCH BREAK	ML	DS	MS
Tue	ML	DS	MS	CPP		**Project LAB @EF CODD**		
Wed	OT	ML	DS	**LIB**		MS	CPP	**LIB/SPROTS**
Thu	DS	ML	CPP	**Seminar**		MS	**Seminar**	
Fri	MS	CPP	ML	DS		**Project Lab**		

Class : IV B.Tech II Semester — w.e.f. 18-11-2019

Sec : D — Class Coordinator : **Mr. M.Vamsikrishna** — Room No : PG-205

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	CPP	DS	ML	MS	LUNCH BREAK	**Project Lab**		
Tue	MS	CPP	ML	DS		ML	CPP	**LIB/SPROTS**
Wed	OT	**Project Lab**				MS	CPP	**Seminar**
Thu	ML	**Project Lab**				MS	DS	**LIB**
Fri	DS	ML	CPP	DS		MS	**Seminar**	

STAFF DETAILS:

S. No.	Course Code	Course Name	Section			
			A	B	C	D
1.		**DS**	Mrs. A. Lakshmi Lavanya	Mrs. A.Leelavathi	Mr. A.Rajesh	Mrs. M Rama Rajeswari
2.		**ML**	Dr. J. Veeraraghavan	Mr. B Krishna Prasad	Dr. J. Veeraraghavan	Mr. B Krishna Prasad
3.		**MS**	Mr. T.Dileep	Mr. T.Dileep	Mr. R.V.RajaSekhar	Mr. K.VinayKumar
4.		**CPP**	Ms. D. SasiRekha	Mrs. D S L Manikanteswari	Mrs. D S L Manikanteswari	Ms. D. SasiRekha
5.		**Seminar**	Dr. V S Naresh	Mrs. A.Leelavathi	Mr. A.Rajesh	Mr. M.Vamsikrishna
6.		**Project Labs**	Mrs. D. SasiRekha	Mr. B Krishna Prasad	Mrs. D S L Manikanteswari	Mr. M.Vamsikrishna
7.		**OT**	Mr. J.N.V. Somayajulu (Mon) Mr.K. Ramana Rao (Thurs)	Mr.K. Ramana Rao (Tue) Mr. T. Suresh (Fri)	Ms. K.V.L.B. Devi (Wed)	Mr. P. SomeswaraRao (Wed)

Head of the Department

COURSE STRUCTURE

IV Year - II Semester

S.No	Course	L	T	P	C
1	Distributed Systems	4	0	0	3
2	Management Science	4	0	0	3
3	Machine Learning	4	0	0	3
4	**Elective-III** i.Concurrent and Parallel Programming ii.Artificial Neural Networks iii. Operations Research	4	0	0	3
5	Seminar	0	3	0	2
6	Project	0	0	0	10
	Total	**0**	**0**	**0**	**24**

Total Course Credits = 48+44 + 42 + 46 = 180

LESSON PLANS

Distributed Systems

Academic Year : **2019-20** **Programme**: B.Tech
Semester : **IV/II** **Sections** :A,B,C&D
Name of the Course: **Distributed Systems** **Course Code**: **412 / R1642051**

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO	Course Outcome	Knowledge Level
C412.1	Describe distributed system and desired properties of such systems.	K2
C412.2	Discuss the basic theoretical concepts of distributed systems and Characteristics of IPC protocols.	K2
C412.3	Demonstrate a simple distributed system.	K3
C412.4	Discuss in detail the system level and support required.	K2
C412.5	Explain the mechanisms such as client/server communication, Remote procedure call (RPC/RMI) and P2P algorithms	K2
C412.6	Discuss modern distributed systems to meet the demands of contemporary distributed applications	K2

TEXT BOOKS:
T1. Ajay D Kshemkalyani, Mukesh Sighal, "Distributed Computing, Principles, Algorithms and Systems", Cambridge
T2. George Coulouris, Jean Dollimore, Tim Kindberg, "Distributed Systems- Concepts and Design", Fourth
Edition, Pearson Publication

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	70
2	60	75
3	60	70
4	60	70
5	60	70
6	60	70

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission, PEOs,POs,PSOs		1	Lecture	BB
2		**Characterization of Distributed Systems:** Introduction about Distributed System	K1	1	Lecture	BB
3		Explain about examples of Distributed System, Resource Sharing and the Web and Challenges	K2	2	Lecture with Discussion	ICT
4		**System Models:** Explain about Architectural Models- Software Layers	K2	2	Lecture	BB
5		Describe about System Architecture and Variations	K2	2	Lecture with Discussion	ICT
6		Describe about Interface and Objects and Design Requirements for Distributed Architecture	K2	2	Lecture	BB
7		Discuss about Fundamental Models, Interaction Model	K2	2	Lecture	BB
8		Discuss about Failure Model and Security Model	K2	1	Lecture	BB
		TOTAL		**13**		

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	**Interprocess Communication:** Introduction The API for the Internet Protocols.	K1	1	Lecture	BB
2		Explain about the Characteristics of Interprocess communication	K2	1	Lecture	BB
3		Discuss about Sockets, UDP Datagram Communication	K2	2	Lecture	BB
4		Discuss about TCP Stream Communication	K2	1	Discussion	ICT
5		Explain about External Data Representation and Marshalling	K2	1	Discussion	ICT+BB
6		Explain about Client Server Communication	K2	2	Lecture	BB
7		Explain about Group Communication	K2	1	Lecture	BB
8		Discuss about IP Multicast an implementation of group communication,	K2	1	Lecture	BB
9		Discuss about Reliability and Ordering of Multicast.	K2	1	Lecture	BB
		TOTAL		**11**		

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	**Distributed Objects and Remote Invocation:** Introduction	K1	1	Lecture	BB
2		Explain about Communication between Distributed Objects-Object Model	K2	1	Lecture	BB
3		Discuss about Distributed Object Model	K2	1	Lecture	BB
4		Discuss about Design Issues for RMI	K2	1	Discussion With Lecture	ICT
5		Discuss about Implementation of RMI and Distributed Garbage Collection	K3	2	Lecture	BB
6		Discuss about Remote Procedure Call	K3	2	Lecture	BB+ICT
7		Discuss about Events and Notifications	K3	1	Lecture	BB
8		Review Case Study: JAVA RMI	K2	1	Lecture	BB
		TOTAL		**10**		

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	**Operating System Support:** Introduction	K1	1	Lecture	BB
2		Discuss about the Operating System Layer and Protection	K2	1	Lecture	BB
3		Distinguish between Processes and Threads and Explain Address Space	K2	1	Lecture	BB
4		Explain about Creation of a New Process	K2	1	Lecture	BB+ICT
5		Explain about Threads	K2	2	Lecture	BB
		TOTAL		**06**		

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	**Distributed File Systems:** Introduction	K1	1	Lecture	BB
2		Explain about File Service Architecture	K2	1	Lecture	BB
3		Discuss about Peer-to-Peer Systems	K2	1	Lecture With Discussion	BB+ICT
4		Describe about Napster and its Legacy	K2	1	Lecture	BB
5		Explain about Peer-to-Peer Middleware	K2	1	Discussion	BB
6		Explain about Routing Overlays	K2	1	Lecture	BB+ICT
7		**Coordination and Agreement:** Explain about introduction of Coordination and Agreement	K2	1	Lecture	BB
8		Explain about Distributed Mutual Exclusion	K2	1	Lecture	BB
9		Discuss about Elections	K2	1	Lecture	BB
10		Explain about Multicast Communication	K2	2	Lecture	BB+ICT
				11		

CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	**Transactions& Replications:** Introduction	K1	1	Lecture	BB
2		Describe about System Model and Group Communication	K2	2	Lecture	BB
3		Explain about Concurrency Control in Distributed Transactions,	K2	1	Lecture	BB
4		Discuss about Distributed Dead Locks	K2	2	Lecture	BB+ICT
5		Explain about Transaction Recovery and introduction of Replication	K2	2	Lecture	BB
6		Distinguish between Passive (Primary) Replication and Active Replication	K2	1	Lecture	BB+ICT
		TOTAL		**09**		

Total No. of Classes: 60

Management Science

Academic Year : 2019-20 **Programme: B.Tech**
Year/ Semester: IV /II **Section:A, B, C, D**
Name of the Course: Management Science **Course Code:C411**

Course Outcomes (Along with knowledge level):
At the end of the course, student will be able to

411.1	Clarify various approaches to Management.[K2]
411.2	Illustrate the principles and practices of operations management.[K3]
411.3	Describes the dynamics of individual and interpersonal behavior in organizational setting through human resource management. [K2]
411.4	Develop the efficient Decision making abilities for the benefit of organization.[K3]
411.5	Explains the better strategic management for organizational effectiveness.[K2]
411.6	Clarify the knowledge of contemporary management practices[K2]

TEXT BOOKS SUGGESTED:
1. Dr. P. Vijaya kumar & Dr. N. Appa rao, management science, cengage.
2. Koontz & weihrich: Essential of Management, TMH 2011

REFERENCE BOOKS SUGGESTED:
1. Philip Kotler & Armstrong: Principles of Management, Pearson publications.
2. Hitt and vijaya kumar: strategic Management, cengage learning.

Targeted Proficiency Level (For each course Outcome):

Course Outcome	Targeted Proficiency Level
CO1	60
CO2	60
CO3	60
CO4	60
CO5	60
CO6	60

UNIT - I

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO1	Definitions of management,	K1	1	Lecture	BB
2.		Describe the Functions of management.	K1	2	Lecture	BB
3.		Evaluation of management thought	K2	2	Lecture+ discussion	BB
4.		Explain Theories of motivation.	K2	2	Lecture	BB
5.		Decision-making process	K2	1	Lecture + discussion	BB
6.		Types of organization structures	K2	1	Lecture + discussion	BB
7.		State the International Management structure.	K2	2	Lecture + discussion	BB
		Total		**11**		

UNIT - II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO2	Describe the Principles and types of management	K1	1	Lecture	BB
2.		Work Study and Statistical Quality Control.	K2	2	Lecture + discussion	BB
3.		Control charts (P- chart, R chart, and C chart)	K3	2	Lecture + discussion	BB
4.		SQC Technic	K2	1	Lecture + discussion	BB
5.		Need for inventory control	K2	2	Lecture + discussion	BB
6.		EOQ, ABC analysis simple problems and Types of ABC analysis.	K3	2	Lecture + discussion	BB
		Total		**10**		

UNIT - III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO3	Concept of HRM, HRD and PMIR	K1	1	Lecture	BB
2.		Functions of HR manager	K2	2	Lecture + discussion	BB

3.		Wage payment plans	K2	2	Lecture + discussion	BB
4.		Job Evaluation and Merit Rating methods.	K2	1	Lecture + discussion and In-class Assignment	BB+PPT
5.		Marketing Management based on product life cycle.	K2	1	Lecture + discussion	BB
6.		Channels of Distributions.	K2	1	Lecture + discussion	BB
7.		Operationlizing change through performance management.	K2	2	Lecture + discussion and In-class Assignment	BB+PPT
		Total		**10**		

UNIT - IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO4	Construction of Network	K2	3	Lecture+ discussion	BB
2.		Difference between PERT and CPM	K2	2	Lecture + discussion and In-class Assignment	BB+ PPT
3.		Compute Critical path, Probability, Project crashing (Simple problems)	K3	4	Lecture + discussion	BB
		Total		**09**		

UNIT - V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	C05	Describe Vision, Mission, Goals and Strategy.	K2	2	Lecture + discussion	BB
2.		Elements of corporate planning process.	K2	2	Lecture + discussion	BB
3.		Environmental Scanning, SWOT analysis.	K2	2	Lecture + discussion and In-class Assignment	BB+ PPT
4.		Explain the Steps in Strategy Formulation and Implementation, Generic Strategy alternatives.	K2	1	Lecture	BB
5.		Global Strategies, theories of Multinational Compnies.	K2	1	Lecture + discussion and In-class Assignment	BB+ PPT
		Total		**08**		

UNIT – VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO6	Explain the Concept ofMIS	K2	1	Lecture + discussion	BB
2.		Describe the concept of MRP,	K2	2	Lecture + discussion	BB
3.		Describe the concept of Just in time.	K2	1	Lecture + discussion and In-class Assignment	BB+ PPT
4.		Describethe concept of Total Quality Management	K1	1	Lecture	BB
5.		Describe the concept of Six sigma	K2	1	Lecture + discussion	BB
6.		Describe the concept of Capability maturity model.	K1	1	Lecture	BB
7.		Describe the concept of Supply chain Management.	K2	1	Lecture + discussion	BB
8.		Describe the concept of Enterprise Resource planning	K2	1	Lecture + discussion and In-class Assignment	BB+ PPT
9.		Describe the concept of Business process out sourcing	K2	1	Lecture + discussion	BB
10.		Explain the concept of business process Re engineering.	K2	1	Lecture + discussion	BB
11.		Describe the concept of Bench marking	K2	1	Lecture + discussion and In-class Assignment	BB+ PPT
12.		Explain the concept of Balance score card.	K1	1	Lecture	BB
		Total		**12**		

Total No. of classes: 60

Machine Learning

Academic Year: 2019-20 **Programme:** B.Tech

Year/ Semester: IV Year II Sem **Section:** A,B,C,D

Name of the Course: MachineLearning

Course Outcomes (Along with Knowledge Level):

Upon completion of the course student will be able to

1	Explain the characteristics of machine learning that make it useful to real-world Problems. **(K2)**
2	Discuss Machine learning algorithms as supervised, semi-supervised, and Unsupervised.. **(K2)**
3	Demonstrate the Tree and Rule based models**(K3)**
4	Illustrate the Linear and Distance based models**(K3)**
5	Demonstrate Regression algorithms..**(K3)**
6	Apply the Neural networks and Dimensionality reduction for non-linear functions **(K3)**

TEXT BOOKS:

1. Machine Learning: The art and science of algorithms that make sense of data, Peter Flach, Cambridge.
2. Machine Learning, Tom M. Mitchell, MGH.

REFERENCE BOOKS:

1. Understanding Machine Learning: From Theory to Algorithms, ShaiShalev-Shwartz, Shai Ben-David, Cambridge.
2. Machine Learning in Action, Peter Harington, 2012, Cengage.

Targeted Proficiency Level (For each course Outcome):

Course Outcome	Targeted Proficiency Level
CO1	60
CO2	60
CO3	60
CO4	60
CO5	60
CO6	60

Targeted Level of Attainment (For each course Outcome):

Course Outcome	Targeted Attainment Level
CO1	60
CO2	60
CO3	60
CO4	60
CO5	60
CO6	60

UNIT-I: The ingredients of machine learning, Tasks

S#	Course Outcomes	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1	CO1	Dissemination of Course Outcomes. Identify the importance of Machine Learning	K1	1	Lecture	BB
2		Illustrate the meaning of the task(problems) that can be solved with machine learning,	K2	2	Lecture + discussion and In-class Assignment	BB & Projector
3		Illustrate the classifying the models in Machine Learning	K2	2	Lecture + discussion	BB & Projector
4		Illustrate Features, and the workhorses of machine Learning.	K2	2	Lecture + discussion	BB& Projector
5		IllustrateClassification, Scoring and ranking	K2	2	Lecture + discussion	BB& Projector
		Total		9		

UNIT-II:Beyond binary classification

S#	Course Outcomes	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO2	Demonstrate Handling more than two classes	K2	2	Lecture + Demonstration	BB & Projector
2		Demonstrate Regression,	K2	2	Lecture + Demonstration	BB & Projector
3		Illustrate Unsupervised and descriptive learning.	K2	1	Lecture + Demonstration	BB & Projector
4		Demonstrate The hypothesis space	K2	2	Lecture + Demonstration	BB & Projector
5		Demonstrate Paths through the hypothesis space,	K2	1	Lecture + Demonstration	BB & Projector
6		Demonstrate Beyond conjunctive concepts	K2	1	Lecture + Demonstration	BB & Projector
		Total		9		

Unit-III:Tree models and Rule models

S#	Course Outcomes	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO3	Demonstrate Decision trees	K3	2	Lecture + Demonstration	BB & Projector
2		Demonstrate Ranking and probability estimation trees,	K3	2	Lecture + Demonstration	BB & Projector
3		Demonstrate Tree learning as variance reduction.	K3	2	Lecture + Demonstration	BB & Projector
4		Demonstrate Learning ordered rule lists,	K3	2	Lecture + Demonstration	BB & Projector
5		Demonstrate Learning unordered rule sets	K3	2	Lecture + Demonstration	BB &Projector
6		Demonstrate Descriptive rule learning	K3	2	Lecture + Demonstration	BB & Projector
7		Demonstrate First-order rule learning	K3	2	Lecture + Demonstration	BB & Projector
		Total		14		

Unit-IV:Linear models and Distance Based Models

S#	Course Outcomes	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO4	IllustrateThe least-squares method	K2	1	Lecture + Demonstration	BB & Projector
2		IllustrateThe perceptron: a heuristic learning algorithm for linear classifiers,	K2	1	Lecture + Demonstration	BB & Projector
3		DemonstrateSupport vector machines	K3	2	Lecture + Demonstration	BB & Projector
4		obtaining probabilities from linear classifiers,	K3	2	Lecture + Demonstration	BB & Projector
5		IllustrateGoing beyond linearity with kernel methods	K2	1	Lecture + Demonstration	BB & Projector
6		IllustrateDistance Based Models:Introduction,	K2	1	Lecture + Demonstration	BB & Projector
7		DemonstrateNeighbors and exemplars,	K3	2	Lecture + Demonstration	BB & Projector
8		Demonstrate Nearest Neighbours classification	K3	2	Lecture + Demonstration	BB & Projector
9		Demonstrate Distance Based Clustering,	K3	2	Lecture + Demonstration	BB & Projector
10		DemonstrateHierarchical Clustering.	K3	2	Lecture + Demonstration	BB &Projector
		Total		16		

UNIT-V:Probabilistic models, Features and Model ensembles

S#	Course Outcomes	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO5	DemonstrateThe normal distribution and its geometric interpretations,	K3	2	Lecture + Demonstration	BB & Projector

2		DemonstrateProbabilistic models for categorical data,	K3	2	Lecture + Demonstration	BB & Projector
3		DemonstrateDiscriminative learning by optimizing conditional likelihood Probabilistic models with hidden variables	K3	2	Lecture + Demonstration	BB & Projector
4		DemonstrateKinds of feature, Featuretransformations, Feature construction and selection.	K3	3	Lecture + Demonstration	BB & Projector
5		DemonstrateModel ensembles: Bagging and random forests, Boosting	K3	2	Lecture + Demonstration	BB & Projector
		Total		11		

UNIT-VI:Dimensionality Reduction and Artificial Neural Networks

S#	Course Outcomes	Contents	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO6	Principal Component Analysis (PCA), Implementation and demonstration	K3	3	Lecture + Demonstration	BB & Projector
2		Illustrate to Introduction to ANN	K2	2	Lecture + Demonstration	BB & Projector
3		Demonstration Neural network representation appropriate problems for neural network learning	K3	3	Lecture + Demonstration	BB & Projector
		Total		8		

Total Hours= 68

Concurrent and Parallel Programming

Academic Year: 2019-20 Programme: B.Tech

Year/ Semester: IV/II Section: A, B,C & D

Course Name: Concurrent and Parallel Programming Course Code: C- 412

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1. Illustrate concurrent and parallel programming. (K2)

CO2. Describe issues Process and threads. (K2)

CO3. Develop Different parallel algorithms. (K3)

CO4. Describe parallel programming Models. (K2)

CO5. Illustrate the multiprocessing and multi-threading programming interfaces.(K2)

CO6. Discuss the Heterogeneous computing techniques.(K2)

Text Books suggested:

T1. Mordechai Ben-Ari. Principles of Concurrent and Distributed Programming, Prentice-Hall International.

T 2. Greg Andrews. Concurrent Programming: Principles and Practice, Addison Wesley.

T 3. GadiTaubenfeld. Synchronization Algorithms and Concurrent Programming, Pearson.

T 4. M. Ben-Ari. Principles of Concurrent Programming, Prentice Hall.

T5. Fred B. Schneider. On Concurrent Programming, Springer.

T 6. Brinch Hansen. The Origins of Concurrent Programming, From Semaphore.

Reference Books suggested:

R1: Matthew J. Sottile, Inttroduction to Concurrency in Programming Languages, CRC Press

R2: Joe Duffy, Concurrent Programming on Windows

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
C- 412.1	60	60
C- 412.2	60	60
C- 412.3	60	60
C- 412.4	60	60
C- 412.5	60	60
C- 412.6	60	60

Unit-1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Generalize Concurrent Programming sequential programming.	K2	1	Lecture	BB
		Explain Concurrent versus sequential programming.	K2	1		
2		Discuss Concurrent programming constructs.	K2	2	Lecture	BB
3		Clarify Concurrent programming race condition.	K2	2	Lecture	BB
4		Describe Synchronization primitives.	K2	2	Lecture	BB+ICT

Unit- 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Classify Processes and threads.	K2	2	Lecture	BB
2		Explain Interprocess communication.	K2	2	Lecture with Discussion	BB

3		Generalize Live lock and deadlocks.	K2	2	Lecture with Discussion	BB+ICT
4		Clarify starvation, and deadlock prevention.	K2	2	Lecture	BB
5		Review Issues and challenges in concurrent programming paradigm.	K2	2	Lecture with Discussion	BB+ICT
6		Expalin concurrent programming paradigm current trends.	K2	2	Lecture with Discussion	BB+ICT

Unit-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**CO3**	Apply Parallel algorithms for sorting.	K3	2	Lecture	BB
2		Find Parallel algorithms for ranking.	K3	2	Lecture with Discussion	BB + ICT
3		Illustrate Parallel algorithms for searching.	K3	2	Lecture with Discussion	BB + ICT
4		Apply Parallel algorithms for traversals.	K3	2	Lecture with Discussion	BB+ ICT
5		Illustrate Parallel algorithms for prefix sum.	K3	2	Lecture with Discussion	BB+ ICT

Unit- 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**CO4**	Discuss Parallel programming paradigms.	K2	1	Lecture	BB
		Classify Data parallel.	K2	1	Lecture	BB
2		Explain Task parallel.	K2	2	Lecture with	BB+ ICT

					Discussion	
3		Extend Shared memory and message passing.	K2	2	Lecture with Discussion	BB+ ICT
4		Estimate Parallel Architectures.	K2	2	Lecture with Discussion	BB+ ICT
5		Explain GPGPU.	K2	2	Lecture with Discussion	BB+ ICT
6		Clarify pthreads.	K2	2	Lecture with Discussion	BB+ ICT
7		Discuss STM.	K2	2	Lecture	BB

Unit-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Illustrate Open Mp.	K2	2	Lecture with Discussion	BB
2		Explain OpenCL.	K2	2	Lecture with Discussion	BB + ICT
3		Express Cilk++.	K2	2	Lecture with Discussion	BB + ICT
4		Illustrate Intel TBB.	K2	2	Lecture with Discussion	BB + ICT
5		Describe CUDA.	K2	2	Lecture with Discussion	BB + ICT

Unit-6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Associate Heterogeneous Computing.	K2	2	Lecture with practical	BB + ICT
2		Express C++AMP.	K2	2	Lecture with practical	BB + ICT
3		Describe Open CL.	K2	2	Lecture with Discussion	BB + ICT

Total No. of Classes: 60